

The required Python libraries are imported to perform agricultural data analysis, machine learning operations, and visualization within the OptiCrop system. Libraries such as NumPy and Pandas are used for numerical computation and dataset handling, while Matplotlib and Seaborn are used for graphical visualization and analytical plotting.

Scikit-learn modules are integrated for machine learning model development, clustering, dataset splitting, and performance evaluation. The visualization style used in this project is FiveThirtyEight (fivethirtyeight), which helps generate clean and professional analytical graphs for agricultural data interpretation and crop prediction analysis.